S P E C T R A V I D E O

U S E R S   G R O U P -- Wgtn

Newsletter Number 12 -- July 1985

The next meeting is on Monday ,July 22th at 7.30 in the Staff Common Room , Wellington Clinical School , Mein Street Newtown. This will be our AGM , an agenda is included with this newsletter.

Wellington SV Users Group

Secretary / Newsletter Editor

Don Stanley Ph 896379 (hm) , 746906 (work)

Treasurer

Colin Chin Ph 325656 (hm)

Committee members

Garry Clark	367872	
Ross Wakelin	793948	
Dennis Timmons	837200	
Darryl Alison	85522 (Paraparaumu) (hm)	399510 (wk)

Specific areas of responsibility for committee members are ... Don -- secretary & newsletter ; Colin -- Treasurer & MSX developments ; Garry -- Radio Access & public domain tape software ; Ross -- National Committee ; Dennis -- library ; Darryl -- program evaluation committee chairman & Kapiti users rep

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Club Subscriptions

Note that expiry date for subscriptions is printed on the address label of newsletter. Subscriptions not renewed a month after this date will not receve newsletters from then.

Number of Members = 146

PROTECTION OF LINES ON THE CRT SCREEN

Illustration by modification of the TEXTPRINT programme

by Graeme Dick

If you don't have a disk drive, 80-column card and word processing programme such as WORDSTAR, and don't have the JUST WRITE JNR cartridge for your cassette-based system, then you will probably have to resort to a locally-produced programme, TEXTPRINT, for simple text editing.

TEXTPRINT allows simple entering and editing of text, but only within the limits of the previously entered line length. In practice this means that once a line is entered it can be edited only to correct typographical errors - it is too cumbersome to make major alterations to a line and then make the necessary changes to all the succeeding lines. This doesn't really matter for many applications, such as writing letters, Minutes of Meetings etc., for which I find the programme quite acceptable.

For me, working in 40-column mode, the most annoying limitation of the programme is that in entering a line it is difficult to judge when the maximum line length has been exceeded. This is found out only after typing the line and pressing <enter>, at which time the message "Line too long: Re-enter" is given. The problem is that the printed line will just about always be longer than the CRT line width, and it will probably be some number of characters between 60 and 70 depending on the margins used.

It would be much easier if you could have an on-screen 'line ruler' to show the line length. The following modification does just that, using a tip from Newsletter No. 5, October 1984. Location FA06 can be used to protect lines at the bottom of the screen - to protect n lines, poke 256-n into that location.

Enter the following subroutine and then make the listed changes to the existing programme so that the subroutine can be called when required:

```
2500 A$="_________"+CHR$(211):B$=""
2510 FOR I=1 TO 4:B$=B$+A$:NEXT
2520 LOCATE0,22:PRINT" Width=40";SPC(13)"Line length=";LL
2530 IF LL<41 THEN LOCATE(LL-1),20:PRINT"X" ELSE IF LL<81
THEN LOCATE (LL-41),20:PRINT"X" ELSE IF LL<121 THEN
LOCATE (LL-81),20:PRINT"X" ELSE IF LL<161 THEN LOCATE
```

```
(LL-121),20:PRINT"X"
2540 LOCATE0,21:PRINTB$:POKE &HFA06,252:LOCATE 0,2:RETURN
```

Alter the following lines by adding what is printed in capitals:

```
450  ....cls:SCREEN,0
590 cls:SCREEN,0:..........
610 cls:SCREEN,0:..........
1309 ...... :GOSUB 2500 (at end of line)
1420 GOSUB 2500:LOCATE 0,8:..........
1608 cls:SCREEN,0
1620 ......... cls:SCREEN,0:....
2311 gosub1609:GOSUB 2500:return
```

The subroutine prints the ruler and also marks the maximum line length on screen. Now, when text is entered and the bottom of the screen is reached, the ruler stays put and the text above it is scrolled.

As you can see, a lot of SCREEN,0 instructions have also been added. This instruction turns off the function key display. These instructions are added because whenever a CLS is done a value of 255 is poked into location FA06 and the function key display is turned back on! You're now back in SCREEN,1 and the protection at the bottom of the screen has been lost.

Incidentally, the way to clear protected lines is to POKE 0 into FA06, and this is also what SCREEN,0 does. I have also discovered that if you POKE an odd number into FA06 the SHIFT key will return the display to SCREEN,1!

PROGRAMMING TIPS

GARTH THORNTON

If you need to save space, it may be useful to use variables for commonly used constants - even small ones. While a 0 or a 1 may look like a single character, several bytes are used to store it in program memory. So your program could start with z=0:a=1:b=2:c=3:d=10:e=100 for example. If you must use 0 as a variable, make it zero - when typing in a large program it is too easy to get a few wrong otherwise.

****

For advanced programmers...

A different method of including small machine code routines in BASIC programs was noticed in an atari program. The code is written as a string, with each byte represented by its ASCII or graphics equivalent. The code does not need to be poked into RAM: this happens automatically when the string is defined. You don't even have to worry about where the routine is in RAM - but of course it must be relocatable (i.e. it doesn't mind where in RAM you put it). To use the routine, use DEFUSR=VARPTR(A$) for a string A$. Unfortunately the Spectravideo character set is not complete, so you would have to use CHR$() for some bytes, i.e. 0-31,127-159,212-255. The limit is 255 bytes (max. size of a single string) although you would need to use 2 program lines for anything much over 100 bytes; the first line A$="...."+chr$(.)+"..."+... etc, the second line A$=A$+"..."+...etc.

Newsletter Contributions

If you are submitting programs to go in the newsletter they MUST satisfy the following conditions or they will NOT be printed.

1 ... be typed/printed on A4 paper with about 5 lines left blank at top and bottom of page and an inch at least around the sides

2 ... typed with a dark ribbon OR printed using emphasized print mode

3 ... include a description of what the program does , typed on A4 paper with example input

4 ... In the case of programs above 20 lines in length they should be sent on cassette , in this case there is no need for the printout as I can merge them into the newsletter. Enclose a stamped addressed envelope for return of cassette

5 ... line numbers should be continuous -- do a RENUM before printing the programs

6 ... PLEASE MAKE SURE THEY WORK . and would readers please note that I am NOT responsible for programs sent by people which do not work .

7 ... In the case of screen dumps from 3d plotters please indicate the function used and try to get the print as dark as possible -- not always easy i know .

All other newsletter contributions should be sent to me at P.O Box 7057 , Wellington South. They may miss being printed for a month if sent to any other box or given to anyone else. Articles should follow 1 and 2 above.

Deadline for Newsletter articles is the 30/31st of each month.

Please follow these guidelines, articles with a 4mm edge all around can not be printed , and neither can articles produced on a dot matrix printer with low resolution . They are simply unreadable after photocopying.

Discounts for members .. All discounts offered by Einstein Scientic and Microstyle Computers do NOT include hardware. Discounts may be offered on such peripherals as floppy disks, cassettes , books , furniture, printers , monitors but NOT on SpectraVideo hardware. Individual dealers may not offer all the goods above at discounted rates , inquire from the shops just what discounts are available to club members. Note that to obtain a discount on any goods you must produce your current membership card.

GAMES SCORES

Spectron	11710	Robert Campin
Sector Alpha	666400	Gavin Knight
Kung Fu	38170	Taane Clark
	17690	Garry Clark (*)
S C Force	748600	Deirdre Chin
Telebunny	87810	Colin Chin
	26470	Don Stanley (+)
Omac	35360	Richard Bremford
	25700	Don Stanley (+)
Ninja	136900	Nicola Tuinman
Tetra	266600	Nicola Tuinman
Turboat	68390	Garth Thornton (*)
Sasa	118240	John Tuinman
Frantic Fred	11900	Don Stanley (level4)
Flipper Slip	29990	Karen Stanley (level1)

(+) = started in normal level
(*) - started in arcade level
otherwise practice level or unknown

Extra Club Meetings ....

The first meeting of our Kapiti subgroup will be on JULY 21st. The venue is Kapiti Community Centre , Hinemoa Street starting at 1.00 .

Also attempts are underway to form a Palmerston North sub-group of our group . If you are interested contact John , ph 722 Shannon .

Club Sponsers

Einstein Scientific -- 177 Willis Street , Wellington

MicroStyle Computers -- 198 Jackson Street , Petone

CLOAD"JUNIOR"

This is the page where youngsters and the young at heart can make their contributions.

```
10 REM
20 COLOR15,0,0:SCREEN1,2:CO=2:R=RND(-TIME)
30 B$= "L4T255O4EDDCO3AARO4EDDO3BAGGRO4EDDCO3A8R64A8R64
       A8R64A8R64A8R64O4L4CCO3BBO4CO3A8R64AR64AR64GG"
40 LOCATE37,170:PRINT"W E   A R E   T H E   W O R L D "
50 LOCATE20,182:PRINT"W E   A R E   T H E   C H I L D R E N"
60 PLAYB$
70 FORX=1TO32
80 READA$
90 S$=S$+CHR$(VAL("&B"+A$))
100 NEXTX
110 FORZ=0TO31
120 SPRITE$(Z)=S$
130 R=INT(RND(1)*15)
140 PUT SPRITEZ,(15+R+V,U),CO,Z
150 CO=CO+1
160 V=V+55
170 IFV>195THENV=0:U=U+20
180 T=T+1:IFT=120THENPLAYB$:T=0
190 IFCO=16THENCO=2
200 NEXT
210 CO=2:V=0:U=0
220 GOTO110
230 DATA 00001111        390 DATA 11100000
240 DATA 00011011        400 DATA 10110000
250 DATA 00011111        410 DATA 11110000
260 DATA 00011110        420 DATA 11110000
270 DATA 00011011        430 DATA 10110000
280 DATA 10001100        440 DATA 01100000
290 DATA 11000011        450 DATA 11000000
300 DATA 11111111        460 DATA 11111111
310 DATA 00111111        470 DATA 11111100
320 DATA 00000111        480 DATA 11100000
330 DATA 00000111        490 DATA 11100000
340 DATA 00000110        500 DATA 01100000
350 DATA 00000110        510 DATA 01100000
360 DATA 00001100        520 DATA 00110000
370 DATA 00001100        530 DATA 00110000
380 DATA 00011100        540 DATA 00111000
```

To save typing and memory replace lines 230 to 540 with 230 to 260 below and replace in line 90 (&B) with(&H)

```
230 DATA 0F,1B,1F,1E,1B,8C,C3,FF
240 DATA 3F,07,07,06,06,0C,0C,1C
245 REM other half
250 DATA E0,B0,F0,F0,B0,60,C0,FF
260 DATA FC,E0,E0,60,60,30,30,38
```

THE SPRITE IN HEXADECIMAL

HOW ABOUT DESIGNING A SPRITE AND SENDING THEM TO THIS PAGE FOR INCLUSION IN A PROGRAM. IF YOU HAVE ANY QUESTIONS I WILL FIND SOMEONE TO ANSWER THEM. HOW ABOUT SENDING IN YOUR PROGRAMS TO THIS PAGE.
Taane CLARK

THE TWO HALVES OF A 16*16 BINARY SPRITE IN WHICH YOU CAN VISUALIZE A PERSON.

Make your sprite in binary and convert to Hex using this program

```
5  REM CONVERTS Binary to HEX for
10 REM one half of a 16 * 16 sprite
20 CLS
30 PRINT"8 digit:binary No.******** + enter"
40 FOR L = 1 TO 16
50 INPUT"binary number &B";B$
60 H$ = HEX$(VAL("&B"+B$))
70 LOCATE27,L:PRINT"= Hex &H ";H$
80 NEXT
90 PRINT:PRINT"FINISHED"
```

Public Domain Software

1) Do NOT send us tapes. We will supply a standard tape which will get round the problem of people sending tapes which are too short.

2) The cost of tapes will be $3.00 each and we will also have a charge of $2.50 to cover handling and return postage. This is charged for each separate lot of copying done , not per tape. At present we have 4 complete tapes.

3) Any of the PD Software may also be broadcast on Radio Access in our 5 minute slot on the first Saturday of each month.

4) The tapes will contain combinations of cassette & disk programs , some may be set up for 80 columns. Adapt them as you wish to suit yourself.

5) Send orders with payment to PDS ,P.O Box 7057 , Wgtn South, Wellington

6) Send us your public domain software for inclusion in the software library.

We are presently negotiating with the local Osborne group to get some more PDS for CPM from the States .

THANKS

Theres a lot of people who work behind the scenes to help get the magazine ready each month. Heres some acknowledgement for the printers at broadcasting who are so prompt with printing when they get the newsletter , to Karen who helps me label them and staple them , to Dennis who posts them out from the hospital and Garry who races around town collecting them and returning them to me. Thanks folks .

And a word of thanks to these people who let our committee turn their residences upside down at committee meetings. Thanks again.

Surveys

Some more questions arising from survey forms ...

----> doesn't advertising cover the cost of the newsletter

NO . We do not ask dealers to pay very much for advertising and are happy to have one dealer paying towards postage costs and the other dealers subsidise the newsletters . Each newsletter costs about a $ to produce and several complimentary copies are sent out also . Do readers consider $10.00 to much for the newsletter?

------> we rely on the newsletter / users group as our only source of information and want to know about all you learn about SV, MSX, extra hardware/peripherals, videotex,software etc

this is exactly what we try to accomplish, particularly for out of town owners. Hopefully the new newsletter editor will continue on in this way

-----> divide the group up into two groups at meetings for advanced/beginners

this is to be brought up at the AGM

------> allow people to air there problems at meetings and perhaps have a short tutorial on various techniques

this has been occurring since meeting #1 for the first part and more recently for part two ?

-----> more information in newsletter on CP/M

again the onus is on more experienced CP/M users to write on this topic, it is very very difficult to get people to write on any topic though , it seems we have a hard core of 5 or 6 people who are happy to write for the newsletters and share ideas , but many more who are not.

------> meetings need to have a better interface for inexperienced users

agreed, please see the AGM agenda further on

------> does anywhere in the world other than NZ & AUS have SpectraVideos

this question arose from the members readings of various overseas magazines. Unfortunately the countries where SV has had the most impact other than here are HOLLAND , ISRAEL and SOUTH AFRICA and non of these countries have regular computer magazines that are seen here. I have had word that the Dutch Distributors put out a small magazine in Holland , we have a member in Holland but have no contact address for either ISRAEL or SOUTH AFRICA. We exchange newsletters with SV-Access (Canada),Montreal SV/MSX users group,Tasmanian Users Group and send complimentary copies to other groups. With the exception of the above groups we get little back from overseas contacts.

Of the 143 questionnaires sent out we got back 34 , ie 24% . Thanks to those who bothered to reply , these will assist us with such things as what to put in the newsletter, format of meetings and so on.

SPECTRAVIDEO IN AUSTRALIA

by Donald Rogers

I have recently returned from a four-week stay in Queensland, where I took the opportunity of making enquiries about Spectravideo computers and those who use them there.

I started with the Downs Computing shop in Toowoomba. One SV728 had come and been sold but more had not yet arrived. I bought a couple of programs written by L Graham, a local SV-user: WRITER ($A15) and FILER (A10).

I bought a copy of the book "Spectravideo ROM BASIC explained" by the Australasian Users' Group. It is an excellent clearly written manual.

The only retailer of SV computers in the whole of Brisbane is the "Computer Connection", in the Logan City Centre, which is quite a distance from the centre of Brisbane. The prices of SV hardware are about the same in Australia as in NZ, while software prices are generally even higher than in NZ.

There I found out that there are two SV-Users Groups in Brisbane. I rang the secretary of the Brisbane Southside Spectravideo Users Group. She told me that the group has about 35 members. The reason for this small number seems to be that the SV computers are not being promoted very vigorously. The BSSUG produces a monthly newsletter. I also rang the president of the group. He told me that one of their members, Brian Parker, had written a program which makes a SV-318 or 328 behave as a MSX computer. This is done by switching out parts of the ROM and using the RAM instead, so that there is just as much memory available as usual.

I made a phone call to Forrest Data Services, in Western Australia. They sell both hardware and software and have some periferals which may be of interest, eg. a printer interface which does not need an expander. It plugs straight in to the expander socket on the back of the computer. It costs $A85 (plus cable). This could be very handy for a tape-based system. They are on the lookout for programs to market, especially educational ones.

FROM 13

```
480 ' SUBROUTINE C
490 IF TP<=0 THEN RETURN
500 PS=TP-1 : GOSUB 680
510 GOSUB 480 : A=-H : B=H : GOSUB 750
520 GOSUB 580 : A=-2*H : B=0 : GOSUB 750
530 GOSUB 380 : A=-H : B=-H : GOSUB 750
540 GOSUB 480
550 GOSUB 720
560 RETURN
570 '
580 ' SUBROUTINE D
590 IF TP<=0 THEN RETURN
600 PS=TP-1 : GOSUB 680
610 GOSUB 580 : A=H : B=H : GOSUB 750
620 GOSUB 280 : A=0 : B=2*H : GOSUB 750
```

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The ZORK assistance line ...

to ensure you don't loose your light source ensure you have the rope with you and use it at the dome ...

Overseas Developments

From the latest MONTREAL users group newsletter comes information about the SV express . The machine has what looks like a fairly standard 328 keyboard , but without a numeric pad. In place of this is the standard MSX cursor pad with very large cursor keys. In addition the machine extends back further and accomodates a 3.5inch disk drive where the 328/318 cartridge slot is found. The magazine says the machine is very impressive . But please note that there is no gaurantee when or if this will arrive here. Also in this magazine was a photo of BONDWELL's new machine , not a portable and with dual 320k disk drives and 2 screens , 1 is evidently dedicated to menus , the system allows simultaneous printing and other processing. This is the BONDWELL-22.

Wanted to Buy

1 ... A single disc drive for the SV601 expander. Contact Brian Collis, Ph 666-919x854(wk)

FROM 9

```
630 GOSUB 480 : A=-H : B=H : GOSUB 750
640 GOSUB 580
650 GOSUB 720
660 RETURN
670 '
680 ' PUSH ROUTINE
690 SP=SP+1 : ST(SP)=PS
700 TP=PS : RETURN
710 '
720 ' POP ROUTINE
730 SP=SP-1 : TP=ST(SP) : RETURN
740 '
750 ' PLOTTING SUBROUTINE
760 LINE (X,Y)-(X+A,Y+B),CO
770 X=X+A : Y=Y+B : RETURN
```

Harry and some other Nelson enthusiasts have recently formed a NELSON SV Users group , so all the best to them . There are now 12 SV User groups in the country.

If you want to look more at Sieroinski's curve and other curves like this try reading some of Martin Gardners MATHEMATICAL PUZZLES and DIVERSIONS books .

Basic Notes -- Select & Print

This is a reprint of an article in newsletter #1 , along with some extra information about these two keys.

Select & Print are known as dead keys. They appear on the keyboard but appear to do nothing when pressed. This is because BASIC handles them in a different way to other keys.

The easiest way to use SELECT & PRINT is by using Input/Output ports to access these keys. Try the following program ...

```
 10 cls
 20 print "Press the Select or Print Key"
 30 y = inp(&h99)
 40 if y=223 then goto 100
 50 if y=239 then goto 200
 60 goto 30
100 print " You Pressed the PRINT key !!":end
200 print " You Pressed the SELECT key !":end
```

There are other ways of using these keys which don't use the port but rather make use of BASICS keyboard matrix which is found at locations FD75 -- FD7F . Lets explore this matrix here and see how it can be used.

Consider the following table.

location/row	BIT	7	6	5	4	3	2	1	0
FD75	(0)	7	6	5	4	3	2	1	0
FD76	(1)	/	.	=	,	'	:	9	8
FD77	(2)	g	f	e	d	c	b	a	-
FD78	(3)	o	n	m	l	k	j	i	h
FD79	(4)	w	v	u	t	s	r	q	p
FD7A	(5)	↑	⇐	]	\	[	z	y	x
FD7B	(6)	←	ret	stop	esc	rg	lg	ctrl	shift
FD7C	(7)	↓	ins	cls	F5	F4	F3	F2	F2
FD7D	(8)	→		prt	sel	caps	del	tab	space
FD7E	(9)	7	6	5	4	3	2	1	0
FD7F	(10)	,	.	/	*	-	+	9	8

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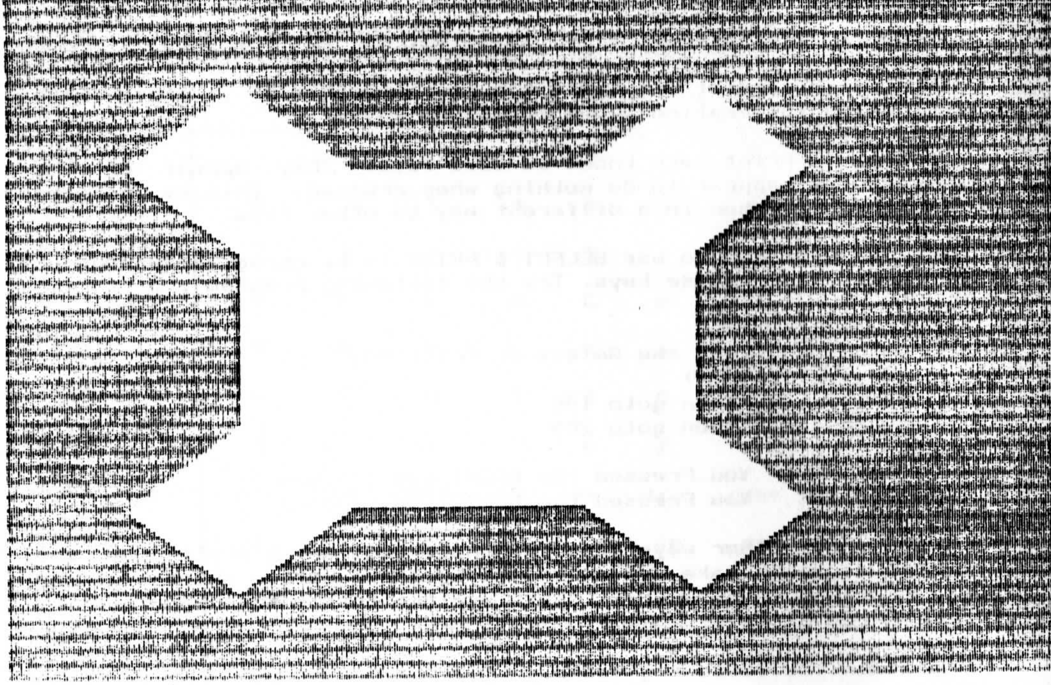

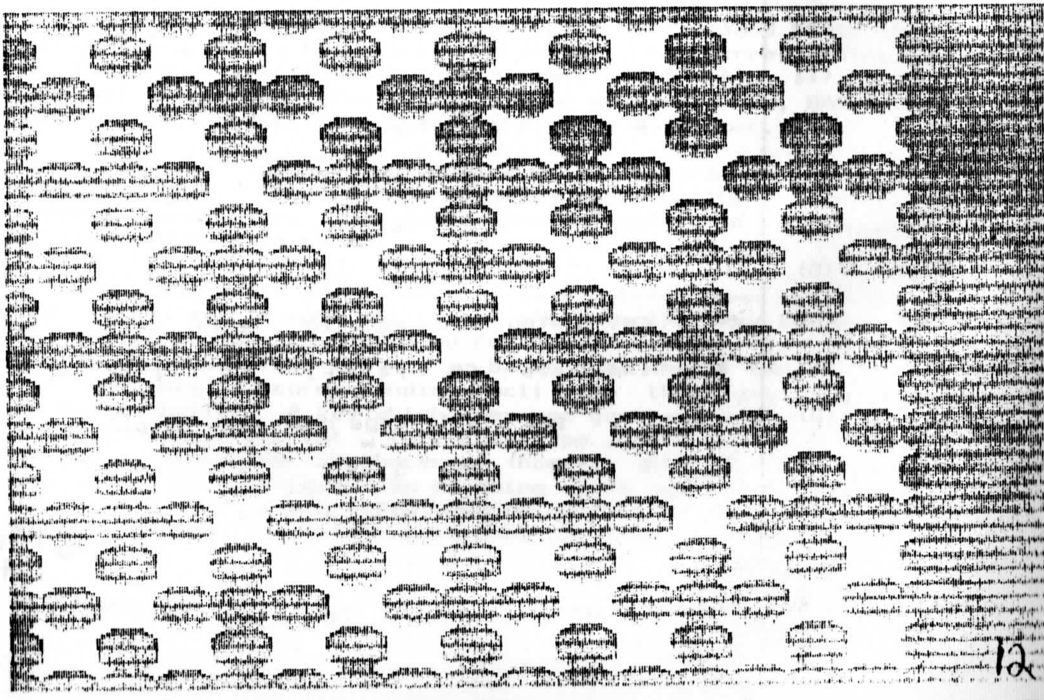

Sierpinski's Curve

The curve on the preceding page is a mathematical curiosity constructed originally by a Polish mathematician , Sierpinski. The curve is unusual in that the top and bottom plots each enclose the same area within the unshaded white part. The top curve is a Sierpinski curve with one part , the bottom has 4. Harry Patterson has written the following program to construct these curves up to 6 parts.

```
10 CLS : COLOR4,4,4 : LOCATE 11,5
20 PRINT "SIERPINSKI CURVE"
50 FOR N=1 TO 900 : NEXT N : CLS
60 INPUT"NUMBER OF CURVES 1 TO 6";A
70 SCREEN 1 : FOR DI=A TO 1
80 N=RND(-TIME)
90 CO=15
100 IF CO=1 THEN 90
110 GOSUB 170
120 PAINT(191,191),CO
130 NEXT DI
140 GOTO 140
150 '
160 ' INITIALISATION AND MAIN CURVE
170 HO=192 : SP=0 : H=HO/4 : X=2*H : Y=3*H : I=0
180 I=I+1 : X=X-H : H=H/2 : Y=Y+H
190 IF I<DI THEN 180
200 PS=I : GOSUB 680
210 GOSUB 280 : A=H : B=-H : GOSUB 750
220 GOSUB 380 : A=-H : B=-H : GOSUB 750
230 GOSUB 480 : A=-H : B=H : GOSUB 750
240 GOSUB 580 : A=H : B=H : GOSUB 750
250 GOSUB 720
260 RETURN
270 '
280 ' SUBROUTINE A
290 IF TP<=0 THEN RETURN
300 PS=TP-1 : GOSUB 680
310 GOSUB 280 : A=H : B=-H : GOSUB 750
320 GOSUB 380 : A=2*H : B=0 : GOSUB 750
330 GOSUB 580 : A=H : B=H : GOSUB 750
340 GOSUB 280
350 GOSUB 720
360 RETURN
370 '
380 ' SUBROUTINE B
390 IF TP<=0 THEN RETURN
400 PS=TP-1 : GOSUB 680
410 GOSUB 380 : A=-H : B=-H : GOSUB 750
420 GOSUB 480 : A=0 : B=-2*H : GOSUB 750
430 GOSUB 280 : A=H : B=-H : GOSUB 750
440 GOSUB 380
450 GOSUB 720
460 RETURN
470 '
```

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These are for the unshifted keys . To obtain the shifted keys , BASIC uses locations FD80 --FD8A as follows ...

location \ BIT	7	6	5	4	3	2	1	0
FD80	&	^	%	$	#	@	!	)
FD81	?	>	+	<	"	;	(	*
FD82	G	F	E	D	C	B	A	_
FD83	O	N	M	L	K	J	I	H
FD84	W	V	U	T	S	R	Q	P
FD85	↑	⇐	}	~	P	Z	Y	X
FD86	←	RET	STOP	ESC	RG	LG	CTRL	SHIFT
FD87	↓	INS	hm	F6	F7	F8	F9	F10
FD88								
FD89			AS FOR FD7D -- FD7F					
FD8A								

In this table the characters in the centre correspond to the characters found in your computer.

OK , now how do these work . Every 50th of a second BASIC scans the keyboard. If a key has been pressed , the corresponding BIT in the above matrix is changed to a 0. (All are 1 if no key pressed). If the shift/ctrl/lg/rg keys were pressed , BASIC then uses the second matrix to obtain the whole character.

For example , suppose you press A which requires the shift key then the a key. The shift is detected and then the a is detected a 50th second later. This would have the effect of setting bit 7 of location FD7B to be 0 , and then bit 6 of FD82 to 0.

If you press shift and nothing , just holding down the shift , the second matrix does not change until another key is pressed, but the first matrix carries on registering the shift. When you lift your finger from the key , basic does one of two things. If the key has a corresponding Ascii code this goes into the keyboard buffer. A key which does not generate an ascii code eg shift , select , print , lg , rg , caps , ctrl is simply ignored as far as the buffer is concerned (however other processing takes place for the caps key). So pressing shift and releasing after a few seconds will cause a few hundred detections of the shift key , but it is forgotten about when the key is lifted (unless another key is pressed of course).

The VDP chip plays an important role here. This chip generates the 50 interrupts per second. Whenever an interrupt occurs the above matrices get completely restored to all 1's. Thus , holding down the shift key actually sees the matrix constantly toggling between byte FD7B being 11111110 and 11111111 .

How do we use this for SELECT and PRINT. Try this ...

```
 10 se = &b11101111  ' SELECT
 20 pr = &b11011111  ' PRINT
 30 print "PRESS SELECT OR PRINT"
 40 x=peek(&hfd7d):if x=255 then 40 ' 11111111b = 255
 50 if x = se then goto 100
 60 if x = pr then goto 200
 70 print "A Key Other Than SELECT or PRINT was pressed"
 80 goto 40
100 print "You Pressed the SELECT key !!":end
200 print "You Pressed the PRINT  key !!":end
```

By the way , if you're wondering how basic detects a key being pressed and changes the relevant bit , the answer is provided by the first part of this article . When a VDP interrupt occurs , port 99 is scanned to read the keyboard and the matrix updated. Of course all this is done in machine code very quickly . It could be done in BASIC as follows ..

```
10 print "Enter Row # ":input row
20 out &h96,row : y=inp(&h99) : if y = 255 then 20
30 print " Row selected was ";row
40 print "Key Pressed was (if ascii key) ";chr$(y)
```

The OUT in line 20 sets up the row of the keyboard to be scanned for input . See the matrix above for the definitions of each row.

It is possible to get BASIC to check for input at some interval other than 1/50 second. POKE &HFE78,n where n is between 0 and 255 sets the keyscan interval which defaults at 1. The effect of this is very noticeable when doing input with the INPUT statement. Poking this results in an effective interval of n/50 second between keyboard scans.

Little Things

CDL have decided not to market either the MSX adapter or the Modem. Little interest has been shown in the modem at 300 baud , while the MSX adapter is already being upgraded by SVI. Hopefully a modem will still be marketed , I know of at least one person working on a modem.

A question which someone may be able to answer ... can pins 5 and 9 on the joystick port be accessed from BASIC and if so --

how ?

And another question -- have any disk users been able to adapt the SVFRMT program so that it exits back to BASIC , rather than rebooting the system. (Incidentally , after loading SVFRMT and formatting and rebooting , SVFRMT still exists in memory and can be rerun using DEF USR=&hC400:A=USR(0) in direct mode)

We have had several requests for copies of the club basic manual and are hopeful that a revised version will be available in a few weeks time.

Notes on Machine Code Programs

A machine code program is a string of bytes which are handled in a completely different way to Basic programs. MC programs CANNOT be listed as can a BASIC program and they are NOT run in the same manner as a BASIC program. Neither are they loaded and saved in the same manner.

Loading a MC program is effectively filling a chunk of memory with a series of bytes. Take SVFRMT for instance, the command BLOAD "1:svfrmt" will load this sequence of bytes into memory. It will NOT run the program.

When a MCP is Bloaded as above where does it go ? This is determined by the first 6 bytes of the file. The first two of these are the (reversed) start location of the program. The next two are the (reversed) end location of the program and the final two are the (reversed) location where the program begins running from. These bytes are set up by the BSAVE instruction.

BSAVE "1:svfrmt",&hc400,&hd500 would cause the segment of memory beginning at &hc400 and ending at &hd500 to be saved to disk as the file svfrmt . Only two parameters are specified , and in this instance the 3rd parameter (where the program runs from) is set the same as the first.

How do we RUN an MCP. Unlike BASIC you don't load the program and then type RUN . There are two ways to start up the program. The first is with the BLOAD command. When you load the program , use BLOAD "1:svfrmt",r which will start the program up immediately. If you do not use the ,r then your program is just a sequence of bytes somewhere in memory , and provided you know the location where the MCP begins to run from you can use DEF USR=position : A=USR(0) to run it.

Bob Curwood writes that the 3rd parameter can be found with the direct statement PRINT HEX$(peek(&FE59))HEX$(peek(&hFE58)). I suspect this means that after bloading you could use the statement

```
Def Usr = VAL("&h"+HEX$(PEEK(&HFE59))HEX$(PEEK(&HFE58)))
A = USR(0)
```

to fire up the routine .

By the way , SVFRMT on the BASIC master disk doesn't need use all the bytes given in the manual to BSAVE with. You can use the statement BSAVE "1:svfrmt",&HC400,&HC70B , however both wili work.

Club Family Evening

Our family evening went off with a bang , indeed a rather unexpected bang as Dennis will testify . 55 club members , wifes , and children gathered at Victoria University to have dinner , see a movie about DONDWELL/SPECTRAVIDEO , find the games champion , and generally have a fun time.

Well, all went very well, and people seemed to enjoy themselves. We lined up 30 youngsters on SASA , FLIPPER SLIPPER , SECTOR ALPHA and FRANTIC FREDDIE , gave them 5 minutes on each games and let them rip. A few problems with SASA being not very co-operative held us up , but then it sure didn't stop ALEXANDER LINDSAY from scoring 27,800 in 5 minutes on that game. Top scorers in the other games were ... JONATHAN MANNING with 8500 for SECTOR ALPHA ; TAANE CLARK with 5680 for FLIPPER SLIPPER and DEIRDRE CHIN with 3810 for FRANTIC FREDDIE. Overall the top 5 were

ALEXANDER LINDSAY	--- 36360 total
JONATHAN MANNING	--- 27880 total
MARCUS MANNING	--- 23760 total
CAMERON CHURCH	--- 20860 total
TAANE CLARK	--- 20500 total

so a copy of one of the latest SV tapes has been sent to ALEXANDER who is happy to take on anyone at all , especially at SASA.

We also began raffling off some of the latest SV software , 4 raffles are running through to the next club night , 2 are $2.00 each (3 titles) and 2 are $1.00 each (one title).

Thanks to evryone who came , it really was an enjoyable evening , and thanks to those who bought computers along and helped make it go with such a 'bang' .

A G M AGENDA

The Annual General meeting of the WGTN SV USERS GROUP will be held as part of the club night on JULY 22nd at Wgtn Clinical School, starting at 7.30 SHARP .

Agenda

1 ... Minutes of last meeting

2 ... Treasurers Report

3 ... Election Of Officers

-- secretary
-- newsletter editor
-- librarian
-- national committe rep
-- treasurer
-- Kapiti representative
-- 1 committe member

Nominations required for all positions , the current office-holders for the last 3 positions (COLIN CHIN,DARRYL ALISON, GARRY CLARK) have indicated they will stand again. The other 4 positions need filling.

4 ... for discussion -- proposal that the group meetings be restructured into a beginners night and an advanced night , each night to be organised 6 weekly rather than monthly with members able to attend either or both nights.

5 ... for discussion -- that the group consider setting a scale of subscriptions so that local members pay $15.00 and out of town members pay $12.00 , and that the extra money be used to cover increasing newsletter costs and costs of meeting place .

6 ... for discussion -- that if 5 is adopted then a suitable definition of an out of town member is one outside the Wellington free dialing area.

7 ... for discussion -- proposal that the user group look into ways of promoting both itself and the SV318/328/728 rather than rely on dealers and distributors to do so.

See you there , we well also have demonstrations of the latest SV BASIC software .

-> 8: General Business

TEXTPRINT

One of the survey forms contained the following information of interest to people who use the TEXTPRINT program.

..' the screen edit facilities of the SpectraVideo greatly simplify the working of the "TEXTPRINT" softwar eby Tom Johnson. Instead of using c/ correction option just use c (change lines) and then move the cursor to the text line and edit it as required, within the line lenght limit, with the screen edit facilities. Pressing "ENTER" records the whole line as it is on screen. This is far simplet than the c/ procedure and much quicker '

Software Review

THINK ... from latest lot of BASIC software

This is a variation of Hangman but with sufficient differences to make it a far more interesting game. All the words have 4 letters , and on each turn you fill in all 4 letters . If a letter is correct in the right place the word is placed in a box with a * beside it (one for each correct letter) and if a letter is correct but in the WRONG place a + goes in the box beside the word. You get about 16 turns to get each word and suitable comments like '.. terrible ... ' are flung back at you when you use lots of turns to get the word. I found this game thoroughly addictive and enjoyable even with hardly any graphics.

GRAPHMASTER ... from latest lot of BASIC software

Enter a function and plot the graph , calculate the integral is what this program does. The graph plotter is slow but accurate and regretably makes no effort to indicate points on the axis. It makes nice use of 'windows' at the side with f key definitons. You can plot multiple functions on one screen. I found the user interface difficult to comprehend ,particularly when entering start and end points on the axis ,and the lack of tickmarks on the axis very confusing, but the graphs are spot on and nicely done.

PARAJUMP ... from latest lot of BASIC SOFTWARE

See review of EMERGENCY LANDING 3 issues ago. Only for those who wish to see how bad commercial programs can get.

JUNGLE JIM ... from latest lot of BASIC software

To big for disk and certainly great fun to play ... rescue

JIMS kidnapped wife and fight off crocodiles and so on on the way. Well worth getting me thinks.

ANTI-METEOR ... from latest lot of BASIC software

Good game to play when the machine code ones have worn your wrists out. You have to stop meteors crashing into the earth , by sliding up and down and shooting you can save the world , but the meteors get faster as they get nearer and why can you only fire once at a meteor until your rocket disappears.? This is a fairly good game but these spacies type games are not suited to basic and get boring very rapidly.

Graphics Routines

From Harry Patterson

```
10 REM MOIRE PATTERNS
20 COLOR 15,1,1 : SCREEN 1 :
   N=RND(-TIME) : DEFINT A-Z : D=191 : E=255 : F=0 : G=2
30 A=(RND(1)*6)+G : X1=RND(1)*E : Y1=RND(1)*D
40 C=(RND(1)*14)+G
50 COLOR C : Y=D
60 FOR X=F TO E STEP A
70   LINE(X,D)-(X1,Y1)
80   LINE-(E-X,F) : NEXT X
90 FOR Y=F TO D STEP A
100   LINE(E,D-Y)-(X1,Y1)
110   LINE-(F,Y) : NEXT Y
120 FOR I=1 TO 1200 : NEXT I
130 CLS: GOTO 30
140 :
150 :
160 '  Routine #2
170 :
180 :
190 :
200 REM NESTED POLYGONS
210 COLOR 15,1,1 : N=RND(-TIME)
220 PI=3.14159
230 C=COS(PI/3) : S=SIN(PI/3)
240 C1=COS(PI/36) : S1=SIN(PI/36) : SF=.95
250 X=95:Y=0 : CX=178 : CY=96 : SC=1.16
260 SCREEN 1 : COLOR RND(1)*14+2
270 FOR J=1 TO 30
280    FOR I=0 TO 6
290      SX=(X*SC+CX)*.7 : SY=CY+Y
300      IF I=0 THEN PSET(SX,SY)
310      LINE-(SX,SY)
320      XN=X*C-Y*S : Y=X*S+Y*C : X=XN
330    NEXT I
340    XN=SF*(X*C1-Y*S1) : Y=SF*(X*S1+Y*C1) : X=XN
```

```
350 NEXT J
360 FOR T=1 TO 800 : NEXT : GOTO 220
370 :
380 :
390 ' Routine #3
400 :
410 :
420 REM MOIRE PATTERNS
430 COLOR 15,1,1 : CLS:N=RND(-TIME) : SCREEN 1 : DEFINT A,B,C,D,X,Y
440 A(1)=RND(1)*255 : B(1)=RND(1)*190
450 A(2)=RND(1)*255 : B(2)=RND(1)*190
460 D=(RND(1)*10)+10
470 C=(RND(1)*14)+2
480 A=A(1) : B=B(1) : GOSUB 520
490 A=A(2) : B=B(2) : GOSUB 520
500 FOR T=1 TO 1000 : NEXT
510 GOTO 430
520 FOR X=0 TO 191 STEP D/4
530   LINE(0,X)-(A,B),C
540   LINE(255,X)-(A,B),C
550 NEXT X
560 REMCO=(RND(1)*14)+2
570 FOR Y= 0 TO 255 STEP D/2
580   LINE(Y,0)-(A,B),C
590   LINE(Y,191)-(A,B),C
600 NEXT Y
610 RETURN
620 :
630 :
640 ' ROUTINE #4
650 :
660 :
670 REM wave form
680 COLOR 15,1,1 : CLS : SCREEN 1 : DEFINT A-Z
690 X= INT(RND(10)*255)
700 Y=INT(RND(10)*175)
710 L=INT(RND(10)*255)
720 M=INT(RND(10)*175)
730 U=15 : V=7
740 DEF FNR(X)=INT(RND(1)*X)
750 GOSUB 930
760 FOR Q=2 TO 16
770   FOR G=1 TO 120
780     NUM=NUM-1
790     IF NUM=0 THEN GOSUB 930
800     LINE(X,Y)-(L,M),(RND(1)*14)+2
810     IF INKEY$=" " THEN GOTO 940
820     IF X+A>255 OR X+A<0 THENA=-A
830     IF Y+B>175 OR Y+B<0 THENB=-B
840     IF L+C>255 OR L+C<0 THENC=-C
850     IF M+D>175 OR M+C<0 THEND=-D
860     X=X+A : Y=Y+B
870     L=L+C : M=M+D
880   NEXT G
890   FOR VV=1 TO 1200 : NEXT VV
```

```
900   CLS
910 NEXT Q
920 IF INKEY$=" " THEN GOTO 940
930 A=FNR(U)-V
940 B=FNR(U)-V
950 C=FNR(U)-V
960 D=FNR(U)-V
970 NUM=FNR(20)+10
980 RETURN
```

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